

ALAN WANG

E-mail: wangan@berkeley.edu Website: <https://urd00m.github.io> github: <https://github.com/urd00m>

EDUCATION:

Ph.D. in Computer Science, University of California, Berkeley	2025-2030
GPA: 4.0 Advisor: Professor Christopher W. Fletcher	
Conjoined BSMS in Computer Science, University of Illinois at Urbana-Champaign	2021-2025
GPA: 4.0 Advisor: Professor Christopher W. Fletcher	

PUBLICATIONS:

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- Pixnapping: Bringing Pixel Stealing out of the Stone Age**, first author, *ACM CCS '25*, <https://www.pixnapping.com>
 - EMT: An OS Framework for New Memory Translation Architectures**, *OSDI '25*
 - Peek-a-Walk: Leaking Secrets via Page Walk Side Channels**, first author, *IEEE S&P (Oakland) '25*
 - Declassiflow: A static analysis for modeling non-speculative knowledge to relax speculative execution security measures**, second author, *ACM CCS '23*

AWARDS/FELLOWSHIPS:

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- NDSEG Fellowship**, DoD, 2026
 - Micro Top Picks Honorable Mention**, IEEE, 2026
 - Jane Street GRF Finalist**, Jane Street, 2026
 - 3x High-Severity Google Bug Bounty Awards**, Google, 2025
 - NSF GRFP Honorable Mention**, NSF, 2025
 - Bronze Tablet Scholar Award**, UIUC, 2025
 - Siebel Scholar Award**, 2025
 - Best Poster in Theme**, SRC PRISM, 2025
 - Dean's List**, UIUC's Grainger College, 2021-2023
 - 2nd Place Overall**, NASA RASC-AL competition, 2022

RECENT EXPERIENCE (see CV for full list):

Research Assistant @ University of California, Berkeley	August 2025 – Current
- Recent work: https://www.pixnapping.com. This work presents a new class of side-channel attacks on Android devices. Most notably, these attacks steal app-based 2FA codes in under 30 seconds - Attacks: researching side-channel attacks inherent to graphics stacks (i.e., pixel stealing attacks) - Defenses: exploring defenses for the side-channel vulnerabilities found by my attack research	
Software Developer Intern @ Jane Street	May – August 2024
Improved the efficiency and utilization of server's used by the entire firm	
Computer Science Intern @ D. E. Shaw Research	May – August 2023
- Researched docking and non-equilibrium FE calculation methods - Wrote system software which ran thousands of simulations on Anton3 ASICs and achieved a 2.5x performance improvement	
Research Assistant @ FPSG Lab	September 2021 – May 2025
- Created a new class of side-channel attacks on Android devices. - Created a new general-purpose cache side channel exploiting page walkers and used it to exploit the Linux kernel using Spectre-V2 and the Intel DDP - Modelled non-speculative information flow to decrease mitigation overhead by up to 21%	
Research Aide @ Argonne National Lab	May 2022 – May 2023
- Worked with NVIDIA's Bluefield-3 Data Processing Unit (DPU) for zero trust network architectures - Found critical errors by instrumenting Portable Batch System (PBS) for Argonne's extreme scale systems - Developed and programmed a command line interface for Argonne's UserBase3 and used by all Argonne admins	

MENTORING/SERVICE:

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- Berkeley Security Seminar Cohost (2026-current)
 - Students mentoring: Jonathan Ji (2026-current), Steven Tang (2026-current), Zaheen Jamil (2026-current), Nathan Vaz (2026-current), Matin Sadeghian (2026-current), Ailsa Sun (2026-current), Kaustubh Khulbe (2025 spring), Ishan Buyyanapragada (2024-2025), Spencer Acquah (2023-2024), Christopher Suarez (2023-2024)
 - ACM Mentor – Helping incoming freshman transition to college life, June 2022 – May 2025

BUGS/NEWS:

CVEs: CVE-2025-48561, CVE-2025-48630

Highlighted news: [Ars Technica](#), [Malwarebytes](#), [Risky Business](#), and [LWN](#)